

Precinct Lineage Across Elections

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Abstract

Election comparisons often assume that a precinct identifier means the same thing across cycles. That assumption fails when precincts split, merge, rename, or move under new reporting conventions. This paper describes how RCOUNT treats reporting-unit lineage as public evidence. A lineage event links prior-cycle units to current-cycle units, records the authority and explanation for the change, and lets multi-election verification distinguish real administrative change from stale plans, broken crosswalks, or tampered sources. Using the `precinct-split-lineage` fixture and the three-cycle synthetic harness, we show how RCOUNT can compare elections without reducing precinct history to fragile string matching.

1 Introduction

Precincts are administrative objects, not eternal atoms. A county may split a precinct after growth, merge two precincts after consolidation, rename a unit, or report vote-center and central-count batches differently from one cycle to the next. A public audit system that compares elections by matching raw precinct labels will eventually confuse administrative change with electoral change.

V.1 described how RCOUNT keeps canvass statuses from overwriting one another. V.2 extends that idea across election cycles. The unit universe for one election is allowed to differ from the next, but the difference must be declared through lineage evidence that a verifier can replay.

The goal is modest and important. RCOUNT does not decide whether a precinct change was wise, lawful, or politically neutral. It verifies that the package declares how prior reporting units relate to current reporting units, that the referenced units exist, that each cycle's sources still hash correctly, and that any RPLAN district aggregation binds the exact unit universe used in that cycle.

2 Lineage Model

An RCOUNT lineage event relates reporting units across cycles. The core fields are:

Field	Meaning
<code>lineage_id</code>	Stable event identifier within the package.
<code>kind</code>	Administrative relation: unchanged, split, or merge in the current draft.
<code>prior_cycle, current_cycle</code>	Package-level cycle labels connected by the event, such as <code>SYN-2024-general</code> .
<code>prior_reporting_unit_ids</code>	Units from the earlier cycle.
<code>current_reporting_unit_ids</code>	Units in the later cycle.
<code>authority</code>	Public office, board, or source asserting the change.
<code>explanation</code>	Human-readable reason, such as growth, consolidation, or re-drawn precinct boundaries.

The verifier checks structure before interpretation. A split must name one prior unit and multiple current units. A merge must name multiple prior units and one current unit. Every referenced unit must exist in the package’s reporting-unit records. Those checks do not prove that the county drew the new units correctly; they prove that the declared cross-cycle relationship is internally replayable. The current crate uses unchanged, split, and merge events; rename and boundary adjustment events are future lineage kinds, not implied by this draft.

Cycle ids are scoped to the package family being compared. They are stable labels for audit transcripts, not national election identifiers. Reporting-unit ids are likewise scoped to the RCOUNT package and its source adapters.

Lineage is distinct from count totals. The lineage event says how units relate; the summaries still carry contest, status, residual, and counted-ballot arithmetic. This separation keeps an administrative crosswalk from silently rewriting votes. The check named `lineage_conservation` conserves the declared unit relationship: it validates split/merge cardinality and referenced unit existence. It does not assert that vote totals are conserved across cycles.

Public source artifacts for lineage can vary by jurisdiction. Examples include precinct-change orders, board records, county GIS releases, public precinct notices, statements of votes, and source crosswalks. RCOUNT normalizes the lineage record and binds public source references where they are available.

3 Split And Merge Fixture

The `precinct-split-lineage` fixture contains two canonical events:

Event	Prior units	Current units
<code>lineage:P-004-split</code>	P-004	P-004A, P-004B
<code>lineage:P-007-P-008-merge</code>	P-007, P-008	P-078

The first event says that P-004 split into two precincts after municipal growth. The second says that P-007 and P-008 merged into P-078 during precinct consolidation. The verifier’s `lineage_conservation` check confirms the cardinality and existence of the referenced units.

Lineage	2024 unit(s)	Relation	2028 unit(s)	Meaning
P-004 split	P-004	$\rightarrow$	P-004A + P-004B	One prior unit becomes two current units.
P-007/P-008 merge	P-007 + P-008	$\rightarrow$	P-078	Two prior units become one current unit.

The fixture is intentionally about evidence shape, not demographic judgment. It does not claim that a split is harmless or that a merge is fair. It gives later analysis a stable, auditable way to avoid comparing P-004 in one cycle to only P-004A in the next.

4 Multi-Election Harness

The L2 synthetic harness spans three cycles: 2024, 2026, and 2028. Each cycle has its own RCOUNT package, RPLAN plan hash, unit universe hash, and district aggregation transcript. The point is not that the synthetic elections are realistic. The point is that every cross-cycle comparison names the exact count package and plan universe that produced it.

Cycle	Units	Lineage events	Unit universe
2024	3	0	P-001, P-002, P-003
2026	4	1	P-001A, P-001B, P-002, P-003
2028	3	1	P-001A, P-001B, P-023

For district aggregation, each cycle binds three hashes:

$$H_{\text{count package}}, \quad H_{\text{rplan plan}}, \quad H_{\text{unit universe}}.$$

In the current transcript, the unit-universe hash binds the sorted reporting unit ids that the aggregation used for that cycle. It is not a substitute for the package hash or plan hash; it is a compact guard that says which unit set the bridge actually saw. That triple prevents a stale crosswalk from being silently reused against a different set of reporting units. If the plan expects P-002 but the current package only contains P-023 after a merge, the failure belongs to the plan/unit boundary, not to vote arithmetic.

This L2 harness is where RCOUNT and RPLAN meet. Base lineage verification does not require RPLAN. District aggregation does, because the district totals depend on a particular plan assignment over a particular reporting-unit universe.

5 Failure Modes

Lineage verification is most useful when failures are specific. The current negative harnesses separate three adjacent problems:

Negative case	Expected failure	Interpretation
bad-lineage	Missing current reporting unit	A lineage event points to a unit that does not exist.
stale-plan	Missing plan unit summary	The RPLAN assignment expects a unit absent from the RCOUNT package.
tampered-2028-source	Source hash mismatch	The package source evidence changed after hashing.

These failures should not collapse into a generic "cross-cycle mismatch." A bad lineage record asks the election package to fix its administrative history. A stale plan asks the district aggregation layer to use the right unit universe. A source hash mismatch asks the package publisher to explain why the evidence bytes changed. The remedy differs because the failure layer differs.

The stale-plan negative is especially important for later district aggregation. It fails before a verifier can safely publish district totals, because the plan assignment refers to a reporting unit that the count package for that cycle does not contain.

6 Verifier Traceability

The lineage examples are executable fixtures. The single-package lineage case is L1 coverage; the three-cycle harness and its negative variants are L2 coverage.

```
cargo run -p rcount-io --example precinct_split_lineage_package
cargo run -p rcount-cli -- verify \
  docs/examples/rcount-golden-packages/precinct-split-lineage

cargo run -p rcount-district --example multi_election_harness
cargo run -p rcount-district --example multi_election_negative_harnesses
```

Fixture	Expected result	Evidence
precinct-split-lineage	pass	Two <code>lineage_conservation</code> checks plus package source hash verification.
multi-election-harness	pass	Three cycles, each with count package hash, RPLAN plan hash, unit-universe hash, and district aggregation checks.
bad-lineage	fail	Lineage event references missing current unit <code>P-023-MISSING</code> .
stale-plan	fail	Plan expects a unit summary absent from the RCOUNT package.
tampered-2028-source	fail	2028 package source hash mismatch.

A compact negative excerpt is enough to show the failure layer:

```
{
  "expected_result": "fail",
  "failure": "core error: lineage event ... references missing current
             reporting unit: syn:precinct:P-023-MISSING"
}
```

The transcript is not saying that P-023 is politically good or bad. It is saying that a lineage event points to an undeclared unit, so the package cannot support that cross-cycle claim.

7 Boundaries

RCOUNT lineage does not prove why voters moved, whether turnout changed, or whether a precinct alteration had political intent. It only states the administrative relationship between reporting units and verifies that the declared relationship is structurally consistent. A passing lineage check is not legal approval of a precinct change.

It also does not require a county to publish voter-level movement records. Lineage works at the reporting-unit level, not the individual-voter level. When future adapters use voter-file, assignment, or registration evidence, they must preserve the V.0 privacy boundary: public verification should not expose voter identity, ballot style, timestamps, or small-cell facts that allow reidentification. Private support evidence can be hashed and retained by an authority without being published; a hash does not sanitize sensitive source data.

Finally, lineage is not a substitute for RPLAN. RCOUNT can declare how reporting units changed; RPLAN declares how those units aggregate into districts for a specific plan and cycle. The bridge between them must remain hash-bound and optional.

RCOUNT lineage enables	RCOUNT lineage does not prove
Replayable split, merge, and unchanged-unit evidence.	Legal validity or political neutrality of a precinct change.
Cycle-specific unit-universe checks.	Individual voter movement.
Failure attribution for bad lineage, stale plans, and source tampering.	That turnout or vote shifts have any particular cause.

8 Conclusion

Precinct lineage makes cross-election count verification honest about administrative change. Instead of pretending that labels are stable, RCOUNT records split and merge evidence, verifies referenced units, and binds multi-cycle aggregation to the count package, plan hash, and unit universe used in each cycle.

This is the foundation for later V-track work. Tamper-evident batch hashing, CVR reconciliation, and district vote aggregation all depend on knowing which public units existed when, how they changed, and which source evidence supports that history.

References

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